

Instance-Based Learning

Based on Tom Mitchell's Machine Learning (Chapter 8)

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1 Introduction

Instance-based learning methods store the training examples and postpone generalization until a query instance must be classified. Instead of constructing an explicit global model, these methods estimate the target function locally for each new instance. [?]

These methods are often called **lazy learning** because most computation occurs during prediction rather than training.

Key Idea

Store training instances $(x_i, f(x_i))$ and use them directly to predict the output for new queries.

2 Lazy vs Eager Learning

Eager learning methods build a model during training.

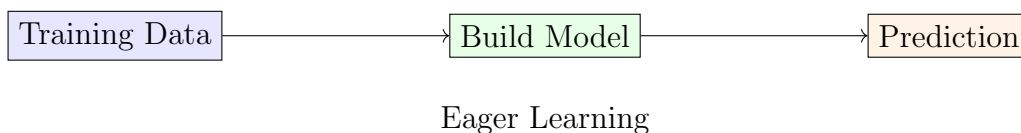
Examples:

- Decision Trees
- Neural Networks
- Naive Bayes

Lazy learning methods delay generalization until prediction time.

Examples:

- Nearest Neighbor
- Case-Based Reasoning



3 Nearest Neighbor Learning

The simplest instance-based learning method is the **Nearest Neighbor algorithm**.

Given a query instance x_q :

1. Find the training example x_n closest to x_q
2. Predict

$$f(x_q) = f(x_n)$$

3.1 Example

Training data:

Instance	Class
(1,1)	+
(2,1)	+
(4,3)	-
(5,4)	-

If a new point $x_q = (2, 2)$ is given, the nearest point is $(2, 1)$, therefore the predicted class is +.

4 k-Nearest Neighbor Learning

Instead of using only one neighbor, the algorithm considers the k closest neighbors.

Prediction rule:

$$f(x_q) = \text{majority vote of } k \text{ nearest neighbors}$$

If the target is numeric:

$$f(x_q) = \frac{1}{k} \sum_{i=1}^k f(x_i)$$

4.1 Algorithm

1. Choose parameter k
2. Compute distance between query and training examples
3. Select k nearest examples
4. Predict using majority vote

5 Distance Metrics

Distance measures determine similarity between instances.

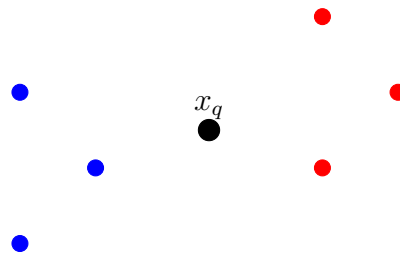
5.1 Euclidean Distance

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^n (x_{ik} - x_{jk})^2}$$

5.2 Manhattan Distance

$$d(x_i, x_j) = \sum_{k=1}^n |x_{ik} - x_{jk}|$$

6 Example Diagram: k-NN Classification



The class of x_q is determined by its nearest neighbors.

7 Locally Weighted Regression

Instead of fitting a global model, locally weighted regression fits a model around the query point.

Prediction:

$$\hat{f}(x_q) = \sum_{i=1}^m w_i f(x_i)$$

Weights depend on distance:

$$w_i = e^{-\frac{d(x_q, x_i)^2}{2\sigma^2}}$$

Closer points receive higher weights.

8 Radial Basis Functions

Radial basis functions are used to approximate target functions.

General form:

$$f(x) = \sum_{i=1}^m w_i \exp\left(-\frac{\|x - c_i\|^2}{2\sigma^2}\right)$$

where

- c_i are centers
- σ controls width

9 Advantages of Instance-Based Learning

- Simple to implement
- No explicit training phase
- Flexible local approximation
- Can adapt to complex target functions

10 Disadvantages

- High memory requirement
- Slow prediction time
- Sensitive to irrelevant features
- Sensitive to noise

11 Summary

Instance-based learning methods:

- Store training examples
- Delay generalization until prediction
- Estimate target function locally

Important algorithms include:

- Nearest Neighbor
- k-Nearest Neighbor
- Locally Weighted Regression
- Radial Basis Functions